

**NODE 1 — ALL-DAY/NIGHT EXTERNAL TUMESCENCE, RIGIDITY, AND PHYSIOLOGY MONITOR**

**A. Overview**

In certain embodiments, a first node is configured for extended wear to measure external tumescence, rigidity-related behavior, and associated physiology over time, including during sleep and daily activity. The first node may be worn continuously for hours and may collect signals suitable for longitudinal pattern measurement, within-user trend comparison, and cross-device correlation with one or more other nodes.

**B. Form Factor and Wear Geometry**

The first node may be implemented as a partially open ring having a C-shape geometry or as a fully enclosed ring. The ring body may comprise a soft external overmold and an internal support frame. The ring may include an inner contact surface configured to maintain contact integrity while minimizing discomfort. The ring body may include one or more flexible segments, variable-thickness regions, relief zones, or padding structures to improve fit stability across a range of anatomies. In certain embodiments, the ring includes an adjustable closure, variable gap region, or size-selection system.

**C. Sensor Suite**

The first node may include one or more sensors and sensing elements, including combinations of the following: 1. Temperature sensing to capture local thermal context and to support wear detection and artifact rejection. 2. Pressure sensing to measure contact force, circumferential loading, and pressure dynamics over time. 3. Strain sensing to measure deformation of the ring body and/or circumferential expansion or contraction patterns. 4. Magnetic sensing using a hall-effect sensor and a magnet to measure relative displacement, gap change, or deformation as a proxy for circumference change. 5. Optical pulse sensing including PPG sensing to capture vascular pulse characteristics and physiologic confirmation. 6. Inertial sensing including a gyroscope and/or IMU to measure motion state, orientation, and to support artifact rejection.

The sensors may be distributed around the ring body, located at one or more points, or arranged in multiple zones. In some embodiments, redundant sensing is used to improve robustness and to generate confidence measures.

**D. Data Collection, Wear Detection, and Signal Quality**

The first node may operate in continuous sampling mode, periodic sampling mode, or event-triggered mode. Wear detection may be performed using one or more of temperature change, contact pressure baselines, impedance or capacitive checks, optical coupling checks, and/or multi-sensor agreement. A signal quality index may be computed to identify invalid segments due to motion, misalignment, poor contact, sensor saturation, or environmental effects. In certain embodiments, inertial sensing is used for motion gating, and contact-based sensing is used to confirm stable measurement windows.

**E. Derived Features and Metrics**

The computing system may derive features from Node 1 signals including circumferential response dynamics, expansion/contraction patterns, duration measures, rate-of-change measures, stability measures, variability measures, and periodicity measures. Metrics may be expressed using privacy-preserving

abstractions, including indices, zones, ranges, and curves. Example metrics include: a tumescence pattern index representing changes over time, a rigidity proxy index based on pressure/strain/hall-effect relationships, a stability index representing fit stability and signal consistency, a duration measure representing time above a threshold or within a zone, and a confidence indicator representing data validity for a given time window.

#### **F. Cross-Device Correlation and Context Conditioning**

Node 1 outputs may be correlated with outputs from an upper-body context node to condition interpretation of patterns based on activity state, heart rate dynamics, temperature context, stress proxies, or sleep context. Node 1 outputs may also be correlated with baseline acquisition results and/or geometry-related outputs from other nodes to normalize measurement and improve interpretation across sessions and time windows.

#### **G. Variations and Alternatives**

The first node may be implemented with different combinations of sensors. For example, certain embodiments omit optical sensing and rely on pressure/strain/magnetic sensing; other embodiments omit magnetic sensing and rely on strain sensing and pressure sensing; other embodiments add additional sensing zones. The ring may be rigid, semi-rigid, or flexible. The node may include local storage, intermittent upload, or real-time streaming. The node may be powered by rechargeable battery, inductive charging, or other power subsystems. The node may include sealing features for sweat and moisture resistance.